



The WASH Benefits and SHINE trials: interpretation of WASH intervention effects on linear growth and diarrhoea

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Child stunting is a global problem and is only modestly responsive to dietary interventions. Numerous observational studies have shown that water quality, sanitation, and handwashing (WASH) in a household are strongly associated with linear growth of children living in the same household. We have completed three randomised efficacy trials testing improved household-level WASH with and without improved infant and young child feeding (IYCF) on stunting and diarrhoea in Bangladesh, Kenya, and Zimbabwe. In all trials, improved IYCF had a statistically significant benefit, but WASH had no effect on linear growth. In observational analyses of data from the control groups of the three trials, baseline sanitation was a strong risk factor for stunting in the study populations, suggesting this frequently reported association might be confounded by unmeasured factors of household wellbeing. WASH interventions reduced diarrhoea in Bangladesh, but not in Kenya or Zimbabwe. Intervention promoters visited participants six times per month in Bangladesh compared with monthly in Kenya and Zimbabwe; a review of the literature shows that virtually all published studies that have reported an effect on diarrhoea through home-based water treatment and handwashing promotion achieved high adherence by visiting participants at daily to fortnightly intervals. Despite achieving substantial behavioural change and significant reduction in infection prevalence for some enteric pathogens, detection of enteropathogens among children in the WASH groups of the trials was typically at ten times higher prevalence compared with high-income countries. Considering these results, we recommend that future research in the WASH sector focus on developing and evaluating interventions that are radically more effective in reducing faecal contamination in the domestic environment than the interventions implemented in these trials.

Introduction

Globally, linear growth faltering (ie, stunting) is the most prevalent form of undernutrition, occurs primarily between conception and 2 years of age, and is associated with higher mortality, reduced educational attainment, and reduced adult economic productivity.¹ Offspring of women who were stunted as children are at greater risk of stunting, creating an intergenerational cycle of low human capital.¹ Although stunting is usually attributed to inadequate diet or disease (especially diarrhoea), this condition has been largely intractable to interventions targeting these factors: a meta-analysis² from 2017 estimated that complementary feeding interventions increase mean length-for-age Z score (LAZ) by 0.11 (ie, less than 10% of the deficit of the median Asian or African child). Similarly, an analysis of observational cohort studies³ estimated that eliminating all diarrhoea in the first 2 years of life would increase child length by 0.38 cm or 0.13 LAZ. Although the World Health Assembly has pledged to reduce stunting by 40% by 2030, the 2013 *Lancet* Nutrition Series on nutrition⁴ estimated that achieving 90% coverage of ten evidence-based nutrition interventions in the 34 highest burden countries would only reduce global stunting by 20%. Thus, we do not have evidence-based solutions for the preponderance of the global child stunting problem.

The 1990 UNICEF framework for undernutrition recommended a multisectoral approach.⁵ Improved water, sanitation, and hygiene (WASH) was highlighted in this framework primarily for its association with diarrhoea. In 2009, we reiterated a hypothesis that

the adverse effects of poor WASH on undernutrition might be partially mediated through diarrhoea, but primarily through environmental enteric dysfunction.⁶ This dysfunction is a subclinical condition of the small intestine characterised by permeability, malabsorption, and systemic and gut inflammation; environmental enteric dysfunction is nearly ubiquitous among people living in impoverished, unsanitary conditions in low-income and middle-income countries.⁶ The hypothesis is supported by 50 years of reports in the animal husbandry literature documenting poor growth accompanied by chronic immune activation among domestic animals reared in dirty and faecally contaminated environments. That this same phenomenon impairs growth among children living in conditions of poor WASH has been hypothesised for decades. Studies from The Gambia showing that failing linear growth of children was not associated with diarrhoea or poor diet, but was strongly associated with biomarkers of gut permeability and microbial translocation provided further support linking poor environmental hygiene to gut health and stunting.⁷ Because the causal pathways linking infant diet and WASH to linear growth are distinct (ie, improving diet increases nutrient intake whereas improving WASH reduces nutrient waste), delivering the two interventions concurrently was hypothesised to have complementary (or even synergistic) effects on growth; this concept has gained momentum in policy and programming efforts.⁸

We have completed three cluster-randomised trials (ie, WASH Benefits Bangladesh,⁹ WASH Benefits Kenya,¹⁰ and the Sanitation, Hygiene, Infant Nutrition Efficacy

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[SHINE] trial¹¹ in Zimbabwe) testing the independent and combined effects of improved infant and young child feeding (IYCF) and improved household WASH on children's linear growth and diarrhoea. Both WASH Benefits trials had seven intervention groups: water chlorination; sanitation; handwashing with soap; combined water chlorination, sanitation, and handwashing with soap (WASH); IYCF (counselling on complementary feeding and provision of small-quantity lipid-paste-nutrient supplement); IYCF combined with WASH; and control (passive in Bangladesh, and passive and active in Kenya). SHINE was a two-by-two factorial trial with four groups: IYCF (counselling on complementary feeding and provision of small-quantity lipid-paste-nutrient supplement), WASH (combined sanitation, water chlorination, and handwashing with soap), IYCF combined with WASH, and standard of care. In all three trials, the WASH interventions were delivered to the household or compound of the index child (the child enrolled in the trial), and the IYCF intervention administered to the index child. None of the trials intervened on water access (around 75% of households in Bangladesh and Kenya and >60% of households in Zimbabwe used water from an improved source; mean walk time to the water source was <3 min in Bangladesh and 10 min in Kenya and Zimbabwe). None of the trials intervened on community-level sanitation coverage (open defecation <7% in Kenya and Bangladesh; around 50% in Zimbabwe); the sanitation intervention primarily moved targeted households from unimproved to improved sanitation in Bangladesh and Kenya, and from open defecation to improved sanitation in Zimbabwe.

Our studies were proof-of-concept trials: our goal was to determine whether our best effort to optimise the nutritional quality of infant diets (IYCF intervention) and to interrupt child faecal–oral exposure (WASH interventions) could improve linear growth of children. The WASH interventions were informed by years of formative research^{12–14} to understand context-specific access to and use of WASH infrastructure, faecal exposure pathways especially for young children, financial barriers to optimal hygiene practices, emotions triggering hygiene behaviours, common understanding about WASH and health, and child health outcomes valued by parents. Behaviour change communication was based on published models of behaviour change theory: the Integrated Behavioural Model for Water, Sanitation, and Hygiene (IBM-WASH) in Bangladesh;¹⁵ the model of planned, motivated, and habitual hygiene behaviour in Zimbabwe;¹⁶ and three models in Kenya (health belief, social cognitive theory, and persuasion theory).¹⁷ Messages, tools, and participatory activities were developed through iterative piloting. Interventions were delivered during home visits by rigorously and repeatedly trained and closely supervised people who lived in, and were respected by, the study communities. In the WASH Benefits Bangladesh trial,⁹ promoters visited households in the intervention groups six times

per month over the entire 2 years of the trial; in the WASH Benefits Kenya trial¹⁰ and the SHINE trial¹¹ in Zimbabwe, household visits promoting behaviour change occurred monthly. The trials^{9–11} freely provided hardware and supplies (ie, improved pit latrines, handwashing stations, soap, chlorine for point-of-use treatment of drinking water, and small-quantity lipid-paste-nutrient supplement packets) so that use would not be constrained by access or affordability; commodities were delivered directly to participant homes with high fidelity across the three trials. Dedicated tools for clearing animal and human faeces from the yard (ie, so-called Sani-Scoops) and potties for young children were provided in the WASH Benefits trials;^{9–10} a washable mat and play yard as a clean play space for the child was provided in the SHINE trial.¹¹ Overall, key indicators of the physical presence of the interventions were moderate to high after 1 year of intervention exposure in the WASH Benefits trials, and at age 1 year in the SHINE trial. In Bangladesh, 79% of target households had detectable chlorine residual in drinking water, 95% had improved latrine, and 90% had soap present at handwashing stations; in Kenya, 41% of households had detectable chlorine residual, 90% had improved latrine, and 77% had soap present at handwashing stations; and in Zimbabwe, 58% had detectable chlorine residual, 99% had improved latrine, and 84% had soap present at handwashing stations. Anthropometric outcomes were measured by research staff who were closely supervised and regularly standardised against a gold standard. The same staff assessed diarrhoea prevalence by 7-day maternal recall (all trials used the WHO definition for diarrhoea).

Trial outcomes

In all three trials, the IYCF intervention modestly but significantly increased the mean LAZ score by 0.13–0.25 (about an eighth of the deficit of the average child aged 18–24 months), which is consistent with decades of nutrition studies.² However, in all three trials the WASH interventions had no effect on linear growth and providing WASH concurrently with IYCF had no additional benefit on linear growth compared with providing IYCF alone. In the WASH Benefits Bangladesh trial, all intervention groups except the water chlorination group yielded a significant relative reduction in diarrhoea of 31–40% (absolute decrease of 1.8–2.5 percentage points from a prevalence of 5.7% in the control group), whereas in the WASH Benefits Kenya (prevalence of 27% in the control group) and SHINE trials (prevalence of 10% in the control group) none of the interventions reduced diarrhoea. In Bangladesh, the combined WASH intervention reduced some indicators of environmental enteric dysfunction (EED) in infants aged 3 months and 14 months, but increased indicators of EED at 28.1 (SD 1.9) months;¹⁸ EED outcomes will be forthcoming from the Kenya and Zimbabwe trials. However, irrespective of those results,

our primary findings show that the WASH interventions implemented in these trials did not improve linear growth along the hypothesised pathway from the environment to the gut to linear growth.

Interpretation and implications for programmes

The null effect of household-level WASH interventions on stunting in our trials is inconsistent with a large body of observational studies. Copious studies from low-income and middle-income populations throughout the world have reported that indicators of household-level WASH are significant predictors of linear growth of children after adjusting for a wide array of other covariates including birthweight, gestational age at birth, maternal height, maternal education, and household wealth. In the past 4 years alone (January, 2015–December, 2018), studies have reported that household access to any latrine or an improved latrine ($n=29$ studies), household use of water from an improved source or treatment of drinking water to make it safer ($n=31$ studies), and adequacy of handwashing by members of the household ($n=16$ studies), were significant independent risk factors for stunting for a child living in that same household (appendix pp 2–3, 8–15). Although some of these studies were secondary analyses and might not have been rigorously controlled, this body of literature includes many classic and highly cited studies;^{19,20} the frequent and consistent finding that poor household WASH conditions are strong independent risk factors for poor linear growth of a child living in that house is notable.

We re-analysed data from the control groups in our three trials, treating these children as a prospective birth cohort. We did regression analyses to identify baseline factors that significantly predicted LAZ score at 18–24 months. In all three trials, children in control households that had an improved latrine at baseline (when their mother was enrolled during pregnancy) were about 0.2 LAZ taller at 18–24 months of age than children in control households that did not have an improved latrine at baseline, after adjusting for all other covariates specified in our individual analysis plans (ie, the difference in LAZ associated with improved latrine ownership was +0.22 [95% CI 0.03–0.40] in Bangladesh, +0.15 [0.02–0.28] in Kenya,²¹ and 0.20 [0.08–0.31] in Zimbabwe). The inconsistency between our observational and experimental findings suggests that the association between baseline latrine ownership and LAZ scores for age 18–24 months in our three study populations was confounded by unmeasured factors of household wellbeing. We suggest that many of the frequently reported associations between household-level WASH indicators and child growth might also be confounded, and that drawing causal inference (ie, improving household WASH will lead to improved child growth) might be misguided.

Although observational studies are subject to confounding, randomised trials can have low external

validity. Our results do not prove that all WASH interventions implemented under all contexts will not improve child growth and might not reduce child diarrhoea. External validity is especially a concern for small, individual trials; we suggest the consistent findings on linear growth across three large-scale trials and on diarrhoea across two trials done in different contexts by different investigative groups increases the likelihood that these findings are relevant to other contexts too. Moreover, proof-of-concept studies are usually delivered with greater implementation fidelity than is feasible by programmes, and people who consent to enrolment might be more responsive to interventions compared with the general population. Thus, external validity might be a concern when protective effects rather than null effects are reported because efficacy trials are likely to produce larger effect sizes than would occur in the general population.

We note that our findings might not be relevant to WASH programmes that achieve higher community-level coverage or sustained duration of intervention adoption. Whether or not these characteristics translate into greater health benefits than the household-level interventions implemented in our trials should be rigorously evaluated in future studies.

Analyses published since the start of our trials have shown a strong association between linear growth of children and community-level open defecation or sanitation coverage.^{22–26} Some trials,^{27–29} but not all trials,^{30,31} testing the effect of total sanitation interventions that are community led have shown improvements in linear growth of children. The growth effect does not seem to be related to achieving a threshold of sanitation coverage: across these five trials,^{27–31} sanitation coverage ranged from 28% to 65%, and the two trials^{28,29} that achieved the lowest coverage reported the highest effect. Perhaps social aspects of community sanitation interventions (eg, community cohesion leading to sharing of resources) rather than reductions in faecal exposure mediate improvements in child growth. Future research should be accompanied by process evaluation to identify whether, where, and how high community-level sanitation coverage maintained over longer durations than in our trials improves child health.

We also found that the effect of the WASH interventions on child diarrhoea varied by site. The 2015 Cochrane reviews of point-of-use water treatment³² and handwashing³³ estimated that these interventions reduce diarrhoea by 25–50%. Consistent with those estimates, in the WASH Benefits Bangladesh trial all interventions except water chlorination significantly reduced diarrhoea by 31–40%. A salient difference in the WASH Benefits Bangladesh trial compared with the African site protocols was the frequency of contact between behaviour change promoters and study participants. Promoters in the WASH Benefits Bangladesh trial visited households in the intervention groups six times per month over the entire 2

See Online for appendix

	Frequency of contact between promoter and participant	Estimator	Effect estimate (95% CI)
Handwashing with soap			
Stanton (1987)	Not reported	Incidence density ratio	0.75 (0.66–0.84)
Haggerty (1994)	Not reported	Relative risk	0.94 (0.85–1.05)
Han (1989)	Daily	Incidence density ratio	0.70 (0.54–0.93)
Luby (2004) and (2005)	At least two times per week	Incidence rate ratio	0.47 (0.35–0.59)
Luby (2006)	At least two times per week	Prevalence ratio	0.59 (0.34–0.88)
Langford (2011)	Daily for 2 weeks, then one to two times per week	Incidence rate ratio	0.74 (0.54–1.01)
Luby (2018)	Six times per month	Prevalence ratio	0.60 (0.45–0.80)
Nicholson (2014)	One time per week	Relative risk	0.79 (0.50–1.23)
Hashi (2017)	One time every 2 weeks (fortnight)	Incidence rate ratio	0.65 (0.57–0.73)
Hartinger (2016)	One time per month	Relative risk	0.78 (0.58–1.05)*
Null (2018)	One time per month	Prevalence ratio	0.98 (0.87–1.09)
Humphrey (2018)	One time per month	Prevalence ratio	1.15 (0.93–1.41)
Water disinfection with chlorination or flocculation			
Doocy (2006)	Not reported	Relative risk	0.12 (0.11–0.13)
Mengistie (2013)	Not reported	Relative risk	0.43 (0.38–0.49)
Kirchhoff (1985)	Daily	Relative risk	1.07 (0.88–1.30)
Austin (1993)	Constantly (exact frequency not reported)	Relative risk	1.01 (0.19–5.39) among children 6–24 months; 1.05 (0.25–4.35) among children 25–60 months
Mahfouz (1995)	Chlorination of 3-month; supply staff checked chlorine detection day after family added chlorine	Relative risk	0.55 (0.30–1.00) diarrhoea-specific clinic visits
Semenza (1998)	Constantly (exact frequency not reported)	Relative risk	0.15 (0.07–0.31)
Luby (2006)	At least two times per week	Relative risk	0.45 (0.20–0.83) chlorine 0.36 (0.10–0.71) flocculent
Jain (2010)	Two times per week	Relative risk	1.12 (0.98–1.28)
Luby (2018)	Six times per month	Prevalence ratio	0.89 (0.70–1.13)
Chiller (2006)	One to two times weekly	Relative risk	0.62 (0.47–0.82)
Handzel (1998) and Sobsey (2003)	One time per week	Incidence rate ratio	0.78 (0.73–0.83)
Quick (1999)	One time per week	Relative risk	0.74 (0.65–0.85)
Crump (2005)	One time per week	Relative risk	0.77 (0.62–0.95) chlorine 0.81 (0.69–0.98) flocculent
Lule (2005)	One time per week	Relative risk	0.80 (0.64–1.00)
Reller (2003)	One time per week	Relative risk	0.74 (0.60–0.92) bleach alone; 0.97 (0.76–1.26) bleach and equipment; 0.79 (0.62–0.99) flocculent alone; 0.74 (0.58–1.94) flocculent and equipment
Opryszko (2010)	Two times in first 3 months of project, then no household visits over following year while social marketing implemented	Incidence rate ratio	0.89 (0.70–1.13)
Boisson (2013)	One time per 2 weeks (fortnightly)	Relative risk	0.99 (0.84–1.17)
Ercumen (2015)	One time per month (but staff visited at mid-point between behaviour change visits to test chlorine in water)	Prevalence ratio	0.64 (0.55–0.73)
Null (2018)	One time per month	Prevalence ratio	1.02 (0.92–1.14)
Humphrey (2019)	One time per month	Prevalence ratio	1.15 (0.93–1.41) at 18 months
Studies are sorted in order of highest frequency of contact between promoters and participants to the lowest. Key studies for interventions are presented. See appendix (pp 4–7, 16–31) for full version of table and citations to all studies included. *Value taken from 2014 publication, ³⁶ which differed from the value used in the 2015 Cochrane meta-analysis. ³³ The incidence rate ratio used in the 2015 Cochrane study was 0.74 (95% CI 0.65–0.85).			
Table: Effect of intervention on diarrhoea for randomised controlled trials of the effect of handwashing with soap and point-of-use water (chlorination or flocculation) disinfection methods done in low-income and middle-income countries			

years of the trial to deliver behaviour change interventions, whereas control group households were contacted only for outcome measurement. In the WASH Benefits Kenya trial and the SHINE trial in Zimbabwe, monthly behaviour

change visits occurred for all households enrolled in the trials. Reporting bias could account for the reduction in diarrhoea measured in the WASH Benefits Bangladesh trial because of more frequent contact in the intervention

compared with control groups,³⁴ but three observations fail to support this hypothesis: reductions in diarrhoea were consistent with objectively measured reductions in *Giardia* spp infections;³⁵ reductions in diarrhoea occurred only during the seasonal monsoons, the periods of presumed highest transmission (ie, biological plausibility);⁹ and interventions did not reduce reported bruising or abrasions (included in the morbidity history as negative control outcomes), a finding that suggests that courtesy bias in child illness symptoms was unlikely to explain the effects.

We reviewed the evidence to date for the effect on diarrhoea of interventions promoting point-of-use water treatment and handwashing in low-income and middle-income countries. We included the studies in the 2015 Cochrane reviews of handwashing³³ and point-of-use water treatment³² on diarrhoea, and with the same search string used in the Cochrane reviews, we identified additional studies published since those reviews. Our purpose was not to recalculate pooled estimates of the intervention effects on diarrhoea, but to examine the characteristics of interventions and their delivery implemented in trials that did and did not achieve significant reductions in diarrhoea. For point-of-use water treatment trials, we limited our review to chlorination (the water treatment intervention implemented in all three of our trials) and flocculent-disinfectant studies, reasoning that the latter is similar to chlorination in terms of behaviour and commodity requirements. A total of 12 handwashing and 23 water treatment (27 comparisons) studies were included. We extracted study design, sample size, study duration, description of the intervention, behaviour change theory applied, free provision of commodities, uptake, description of the person promoting the behaviour change and their frequency of contact with the participant, description of the person measuring diarrhoea outcome and frequency of measurement, and effect of the intervention on diarrhoea (appendix pp 4–7, 16–31).

In summary, we observed that virtually all the evidence that promotion of handwashing and point-of-use water treatment with chlorine or flocculent disinfectant reduce diarrhoea come from studies that had daily to fortnightly contact between the behaviour change promoter and study participant (table). Although not all studies targeted long-term habit adoption, data collected after the research interventions promoting handwashing,^{37,38} point-of-use water chlorination, or flocculent-disinfectant^{39–41} ended indicated that behaviours often steeply declined, and the effect on child diarrhoea disappeared. WASH publications regularly report diarrhoea outcome ascertainment frequency, but sometimes do not report contact frequency between behaviour change promoters and participants. Moreover, contact frequency has not been extracted in systematic reviews,^{32,33,42} which suggests that the relationship between sustained daily to fortnightly behaviour change promotion and health effects might not be widely recognised.

Some evidence exists that public intervention activities might increase (and in one study⁴³ possibly sustain) handwashing rates during epidemics of Ebola, pandemic influenza, and cholera.^{43–45} However, outside of these situations, studies that have achieved a high enough adherence to handwashing and point-of-use water treatment to result in reductions in diarrhoea have typically implemented sustained daily to fortnightly contact between promoters and participants. The implications for programming are that point-of-use water treatment with chlorine or flocculent disinfectant and handwashing promotion programmes implemented through sporadic message delivery by health workers, environmental technicians, or social marketing, might not reduce diarrhoea, at least among children younger than 2 years, who have the highest diarrhoea prevalence. We also recommend that WASH evaluations report the intensity of promotion achieved during the intervention period (frequency of contact between promoters and study participants) to guide future programme planners on the level of intervention intensity required to yield health effects.

Despite high implementation fidelity and uptake of interventions, the children who received the WASH interventions in our trials still experienced very high enteropathogenic exposure and infection, showing that environmental faecal contamination remained pervasive despite these interventions. In the WASH Benefits Bangladesh trial, the combined WASH and water chlorination interventions reduced the prevalence of *Escherichia coli* in stored drinking water and in food, but about 50% of the stored drinking water and 40% of the food still had detectable *E coli*, and none of the WASH interventions reduced *E coli* on the caregivers' or children's hands, ambient water, courtyard soil, or sentinel toys.^{46,47} The combined WASH and handwashing interventions significantly reduced *Giardia* infection by 25%; however, 26–30% of children who received those WASH interventions were still infected with *Giardia*.³⁵ The combined WASH and water chlorination interventions reduced hookworm infection by 29–33%, but about 6% of children who received those WASH interventions were still infected with hookworm.⁴⁸ Finally, the geometric mean *E coli* concentration in soil was more than 120 000 most probable number per gram, animal faecal contamination was extensive,⁴⁹ and object and hand mouthing was frequent among young children.⁵⁰

In the WASH Benefits Kenya trial, the water chlorination intervention reduced the prevalence and concentration of *E coli* in stored drinking water, but two-thirds of water samples still had detectable *E coli* and none of the interventions reduced *E coli* levels on the children's hands or toys (unpublished). The combined WASH and water chlorination interventions reduced *Ascaris* infection by 18–22% but 18% of children who received the WASH interventions were still infected with *Ascaris*.³¹ None of the WASH interventions reduced

Giardia infection, which was prevalent (around 39%) in all treatment groups. Finally, 39% of mothers reported that their child had consumed soil in the previous week.⁵¹

In the SHINE trial in Zimbabwe, the WASH intervention reduced maternal-reported child geophagia by 63% but 27% of children in the WASH group still ingested faecal-contaminated soil. The WASH intervention had a very small effect on total parasite infections but had no effect on carriage of bacterial or viral enteropathogens using molecular diagnostics.⁵²

WASH interventions are often evaluated by indicators of process (eg, latrines constructed), behaviour change (eg, presence of soap and water at a handwashing facility, detectable chlorine in drinking water), or exposure (eg, faecal indicator bacteria in water or on hands). Our trials suggest that interventions can succeed in changing these indicators even while substantial enteropathogenic infection ensues. The WASH Benefits Bangladesh trial is particularly important: intervention intensity was very high, water was available on-plot, and reductions in diarrhoea and interruption in enteric pathogen transmission were documented. Yet even with these achievements, more than 95% of children in the intervention groups had *E coli* contamination on their hands (compared with 5% prevalence in US day care centres)⁵³ and more than 50% of stored drinking water had *E coli* contamination (the international guidelines for drinking water quality are 0 colony-forming units of *E coli* per 100 mL water). Across the three trials, enteric infections in intervention groups were typically at ten times higher prevalence than high-income countries. For example, more than 25% of children in the intervention in Bangladesh, 40% of children in the intervention in Kenya, and 29% of children in the intervention in Zimbabwe were infected with *Giardia* compared with reported prevalence of under 2% in the UK.⁵⁴

New approaches to providing WASH services for rural populations in low-income and middle-income countries that are far more effective in completely separating animal and human faeces from living environments are urgently needed. Moreover, future WASH research and programme evaluations should not measure success on the basis of process, behaviour change, or exposure indicators but must also incorporate direct measurement of enteropathogen infection. Methods such as molecular-based multiplex assays to detect a broad panel of specific enteric pathogens⁵⁵ can objectively assess the primary pathway through which WASH interventions improve human physical health (ie, interrupting faecal–oral transmission) and also identify subclinical and co-infections (emerging as important causes of stunting and other adverse child health outcomes). The MapSan trial⁵⁶ in Mozambique is an excellent example of this strategy; the primary trial outcome is combined prevalence of selected enteric infections. Serological methods for analysing antibodies to enteropathogens from dried blood spots (which are low cost and do not

require cold chains for field collection) are also becoming available.⁵⁷

Conclusion

The WASH Benefits and SHINE trials provide evidence that elementary household-level WASH interventions (ie, improved pit latrine, handwashing stations unconnected to a water source, and point-of-use water chlorination) are unlikely to improve child growth, and that providing these interventions concurrently with complementary feeding interventions is unlikely to improve child growth more than the feeding intervention alone. Given the biological plausibility and large body of studies that have reported that these same household-level WASH interventions were strongly associated with child growth, these findings have been a surprise. These interventions are probably not effective enough in reducing faecal–oral transmission of pathogens to support child growth and optimal health. Even when adherence was high enough to reduce diarrhoea (the WASH Benefits Bangladesh trial) and some enteric infections (all three trials), children who received these interventions in our trials still had high faecal exposure and typically ten times higher rates of enteric infection compared with children in high-income countries.

Our trials also show that the elementary household-level WASH interventions implemented in our trials might not reduce child diarrhoea. In particular, evidence that handwashing and point-of-use water chlorination interventions reduce diarrhoea comes from efficacy trials (including the WASH Benefits Bangladesh trial) that achieved very high levels of adherence by employing frequent contact between promoters and study participants. To our knowledge, all published trials of handwashing promotion and point-of-use water chlorination that had less than fortnightly contact have not found reductions in diarrhoea—even when interventions were contextually appropriate, and soap and chlorine were delivered free of charge to participants' doorsteps, as in the WASH Benefits Kenya trial and the SHINE trial. These results do not mean that handwashing and point-of-use water chlorination interventions will only reduce diarrhoea if implemented with intense interpersonal contact; rather, our trial findings highlight that to date, high frequency contact is the only means through which high enough adherence to handwashing and point-of-use water chlorination has been achieved to reduce diarrhoea. Because this level of contact is not feasible for WASH programmes at scale, handwashing and point-of-use water chlorination programmes are unlikely to reduce child diarrhoea unless alternative innovative means of achieving very high adherence are identified that are feasible for programme implementation.

WASH programmes provide many benefits including security, time savings, dignity, and gender equity. However, findings from our trials indicate that even rigorously implemented WASH programmes cannot be

assumed to achieve the child physical health benefits they are usually designed to address: child health benefits can only be claimed if they are measured. We recommend that evaluations of WASH programmes do not rely solely on process, behaviour change, or exposure indicators, but also incorporate direct measurement of enteropathogenic infection.

Findings from our trials provide an urgent call for research to identify interventions that radically reduce faecal contamination in the household environment of low-income and middle-income countries. We term this transformative WASH. Transformative WASH might require high community coverage of improved sanitation facilities,⁵⁸ complete separation of animal faeces from people's living environments,^{12,49} continuous and convenient access to uncontaminated water,⁵⁹ reductions in faecal contamination on surfaces where young children crawl and play,⁴⁶ new technologies to deliver WASH services which might differ from those implemented in high-income countries,⁶⁰ or different modalities of behaviour change.⁶¹ Regardless of chosen technologies and interventions, transformative WASH will certainly require strengthened support for governance systems of financing, operations, monitoring, evaluation, and regulation.

Contributors

AJP and JHH led the drafting, prepared the tables, and prepared the supporting information. JHH conceptualised the consensus project and supervised the literature reviews. All authors contributed to interpretation and writing of the paper.

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